

SOC/L1 - 4.3. Unified Kill Chain

Task 1 - Introduction

The **Unified Kill Chain (UKC)** is an advanced framework that combines concepts from the **Cyber Kill Chain**, **MITRE ATT&CK**, and other threat models to provide a more complete view of modern cyberattacks.

Task 2 - What is a "Kill Chain"

Originating from the military, a "Kill Chain" is a term used to explain the various stages of an attack. In the realm of cybersecurity, a "Kill Chain" is used to describe the methodology/path attackers such as hackers or APTs use to approach and intrude a target. Unlike the traditional Cyber Kill Chain, which mainly focuses on **malware delivery and external attacks**, the Unified Kill Chain covers a wider range of attacker activities, including **reconnaissance, social engineering, exploitation, persistence, privilege escalation, lateral movement, and data exfiltration**.

Answer the questions below

| Where does the term "Kill Chain" originate from? Military

Task 3 - What is "Threat Modelling"

Threat modelling is a cybersecurity process used to identify risks, analyze vulnerabilities, and improve the security of systems and applications. The main steps include:

1. **Identify assets:** Determine which systems, applications, and data need protection and their importance to the organization.
2. **Assess vulnerabilities:** Identify weaknesses and understand how attackers could exploit them.
3. **Create security plans:** Develop actions to reduce or eliminate identified risks.
4. **Prevent future issues:** Implement security policies, secure development practices, and user awareness training.

Threat modelling helps organizations understand their **attack surface**, prioritize risks, and create effective security controls. Frameworks like the **Unified Kill Chain (UKC)** support threat modelling by helping identify possible attack paths and attacker techniques.

Answer the questions below

What is the technical term for a piece of software or hardware in IT (Information Technology?) Asset

Task 4 - Introducing the Unified Kill Chain

The UKC describes **18 phases of a cyberattack**, covering the entire attack lifecycle, from **reconnaissance and initial access to data exfiltration and attacker motivations**. Compared to traditional kill chain models, the UKC is:

- More **modern and detailed**.
- Provides a broader view of attacker behavior.
- Helps defenders understand complex attack paths and improve threat modelling.

The Unified Kill Chain



1	Reconnaissance	Researching, identifying and selecting targets using active or passive reconnaissance.
2	Weaponization	Preparatory activities aimed at setting up the infrastructure required for the attack.
3	Delivery	Techniques resulting in the transmission of a weaponized object to the targeted environment.
4	Social Engineering	Techniques aimed at the manipulation of people to perform unsafe actions.
5	Exploitation	Techniques to exploit vulnerabilities in systems that may, amongst others, result in code execution.
6	Persistence	Any access, action or change to a system that gives an attacker persistent presence on the system.
7	Defense Evasion	Techniques an attacker may specifically use for evading detection or avoiding other defenses.
8	Command & Control	Techniques that allow attackers to communicate with controlled systems within a target network.
9	Pivoting	Tunneling traffic through a controlled system to other systems that are not directly accessible.
10	Discovery	Techniques that allow an attacker to gain knowledge about a system and its network environment.
11	Privilege Escalation	The result of techniques that provide an attacker with higher permissions on a system or network.
12	Execution	Techniques that result in execution of attacker-controlled code on a local or remote system.
13	Credential Access	Techniques resulting in the access of, or control over, system, service or domain credentials.
14	Lateral Movement	Techniques that enable an adversary to horizontally access and control other remote systems.
15	Collection	Techniques used to identify and gather data from a target network prior to exfiltration.
16	Exfiltration	Techniques that result or aid in an attacker removing data from a target network.
17	Impact	Techniques aimed at manipulating, interrupting or destroying the target system or data.
18	Objectives	Socio-technical objectives of an attack that are intended to achieve a strategic goal.

Benefits of the Unified Kill Chain (UKC) Framework	How do Other Frameworks Compare?
Modern (released in 2017, updated in 2022).	Some frameworks, such as MITRE's were released in 2013, when the cybersecurity landscape was very different.
The UKC is extremely detailed (18 phases).	Other frameworks often have a small handful of phases.
The UKC covers an entire attack - from reconnaissance, exploitation, post-exploitation and includes identifying an attacker's motivation.	Other frameworks cover a limited amount of phases.
The UKC highlights a much more realistic attack scenario. Various stages will often re-occur. For example, after exploiting a machine, an attacker will begin reconnaissance to pivot another system.	Other frameworks do not account for the fact that an attacker will go back and forth between the various phases during an attack.

Answer the questions below

In what year was the Unified Kill Chain framework released? 2017

According to the Unified Kill Chain, how many phases are there to an attack? 18

What is the name of the attack phase where an attacker employs techniques to evade detection? Defense Evasion

What is the name of the attack phase where an attacker employs techniques to remove data from a network? Exfiltration

What is the name of the attack phase where an attacker achieves their objectives? Objectives

Task 5 - Goal: In (Initial Foothold)

The **Unified Kill Chain (UKC)** focus on how attackers **gain access, establish persistence, and avoid detection** within a target environment.

Reconnaissance: Attackers gather information about the target using passive and active methods. They look for:

- Systems, services, and vulnerabilities.
- Employee information for phishing or impersonation.
- Credentials and network details for future attacks.

Weaponization: Attackers prepare the infrastructure and tools needed for the attack, such as:

- Command and Control (C2) servers.
- Malware payloads.
- Exploit tools and delivery mechanisms.

Social Engineering: Attackers manipulate users to perform actions that help the attack, such as:

- Opening malicious attachments.
- Entering credentials into fake websites.
- Providing access through impersonation.

Exploitation: Attackers abuse vulnerabilities to execute code on a target system, for example:

- Uploading a reverse shell.
- Exploiting web application vulnerabilities.
- Abusing scripts or software flaws.

Persistence: Attackers maintain access after compromise by:

- Creating malicious services.
- Installing backdoors.
- Connecting systems to C2 infrastructure.

Defence Evasion: Attackers attempt to bypass security controls such as:

- Firewalls.
- Antivirus solutions.
- Intrusion Detection Systems (IDS).
- Web Application Firewalls (WAF).

Command & Control: Attackers establish communication with compromised systems to:

- Execute commands remotely.
- Steal credentials and data.
- Control infected machines.

Pivoting: Attackers move from a compromised system to other internal systems that are not directly accessible from the internet, allowing them to:

- Expand access.
- Reach valuable resources.
- Perform lateral movement.

Answer the questions below

| What is an example of a tactic to gain a foothold using emails? Phishing

Impersonating an employee to request a password reset is a form of what?
Social Engineering

An adversary setting up the Command & Control server is what phase of the Unified Kill Chain? Weaponization

Exploiting a vulnerability present on a system is what phase of the Unified Kill Chain? Exploitation

Moving from one system to another is an example of? Pivoting

Leaving behind a malicious service that allows the adversary to log back into the target is what? Persistence

Task 6 - Goal: Through (Network Propagation)

The **Through** phase begins after an attacker gains an initial foothold in a network. The goal is to **expand access, gain higher privileges, and move deeper into the environment** until the attacker can achieve their objective.

Pivoting: Attackers use a compromised system as a **staging point** to communicate with the victim network, distribute malware, and access internal resources.

Discovery: Attackers gather information about the environment, including:

- Users and permissions.
- Applications and software.
- Files, directories, and network shares.
- System configurations and network details.

Privilege Escalation: Attackers attempt to gain higher-level access by exploiting:

- Vulnerabilities.
- Misconfigurations.
- Weak permissions.

Common targets include:

- SYSTEM/Root privileges.

- Local administrator accounts.
- Privileged user accounts.

Execution: Attackers run malicious code through the compromised system using:

- Remote Trojans.
- C2 scripts.
- Malicious links.
- Scheduled tasks.

Credential Access: Attackers steal usernames and passwords using techniques such as:

- Credential dumping.
- Keylogging.

Using valid credentials helps attackers avoid detection.

Lateral Movement: With stolen credentials and elevated privileges, attackers move across the network to access additional systems and reach their final objectives.

Answer the questions below

As a SOC analyst, you pick up an alert pointing to failed logins from an administrator account. What phase of the Unified Kill Chain would an attacker be seeking to achieve?

Privilege Escalation

Mimikatz, a known post-exploitation tool, was detected on the IT Manager's workstation. The Security logs show that the tool was attempting to dump OS and user secrets. Which Unified Kill Chain phase does this activity correspond to?

Credential Access

Task 7 - Goal: Out (Action on Objectives)

The **Out** phase represents the final stage of an attack, where adversaries have gained access to critical assets and execute their main objectives. These actions typically target the **CIA Triad (Confidentiality, Integrity, and Availability)**.

Collection: Attackers gather valuable information from compromised systems, such as:

- Files and drives.
- Browser data.
- Emails.
- Audio and video data.

This stage primarily impacts **confidentiality**.

Exfiltration: Attackers steal collected data by:

- Compressing and encrypting files.
- Using previously established **C2 channels** to transfer data while avoiding detection.

Impact: Attackers disrupt or damage systems to affect **integrity and availability** through:

- Data destruction.
- Disk wiping.
- Ransomware encryption.
- Website defacement.
- Denial-of-Service (DoS) attacks.

Objectives: The attacker's final goal depends on their motivation, such as:

- **Financial gain:** Deploying ransomware and demanding payment.
- **Espionage:** Stealing sensitive information.
- **Reputation damage:** Leaking confidential data publicly.
- **Operational disruption:** Damaging business processes.

Answer the questions below

While monitoring the network as a SOC analyst, you observe a big traffic spike. Most of the network traffic is sent to an unknown, suspicious IP address. What Unified Kill Chain phase could describe this activity?

Exfiltration

Personally identifiable information (PII) has been released to the public by an adversary. Your organisation is facing reputational losses and scrutiny for the breach. What part of the CIA triad would be affected by this action?

Confidentiality

Task 8 - Practical

Deploy the static site attached to the task. You will need to match the various actions of an attacker to the correct phase of the Unified Kill Chain framework to reveal the flag.

Answer the questions below

Match the scenario prompt to the correct phase of the Unified Kill Chain to reveal the flag at the end. What is the flag?

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